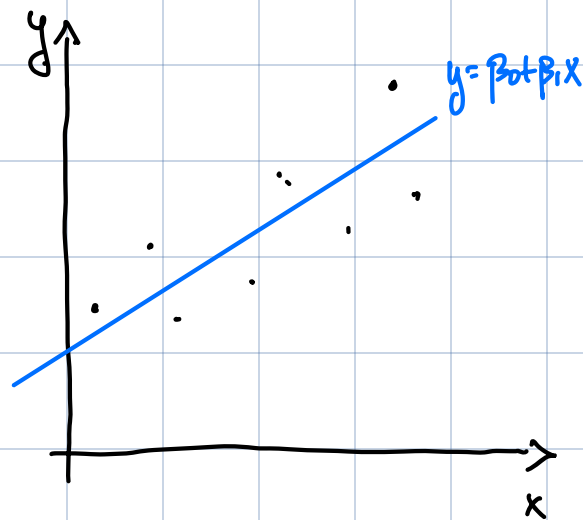


I. Recap of SLR

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j$$



$$\underset{\beta_0, \beta_1}{\text{argmin}} \sum_i (y_i - \hat{y}_i)^2$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{SSX}\right)\right)$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{SSX}\right)$$

$$\hat{y}_i \sim N\left(\beta_0 + \beta_1 x_i, \sigma^2 \left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{SSX}\right)\right)$$

$$e_i \sim N\left(0, \sigma^2 \left(1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{SSX}\right)\right)$$

$$\hat{\sigma}^2 = \frac{\sum_i e_i^2}{n-2}$$

Now do hypothesis tests and confidence intervals
for $\beta_0, \beta_1, E(y)$

$$SST = SSE + SSR$$

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

$$SSR = \sum_i (\hat{y}_i - \bar{y})^2$$

$$SST = \sum_i (y_i - \bar{y})^2$$

$$SSR/\sigma^2 \sim \chi_1^2$$

$$SSE/\sigma^2 \sim \chi_{n-2}^2$$

$$\frac{SSR/1}{SSE/n-2} \sim F_{1, n-2}$$



$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p-1} \\ \vdots & \cdot & \cdot & & \cdot \\ 1 & \dots & \dots & & \dots \end{pmatrix}$$

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_{p-1} \end{pmatrix}$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

II

Matrix Vector Notation

$$\begin{pmatrix} y_1 = \beta_0 + \beta_1 x_1 + \varepsilon_1 \\ y_2 = \beta_0 + \beta_1 x_2 + \varepsilon_2 \\ \vdots \\ y_n = \beta_0 + \beta_1 x_n + \varepsilon_n \end{pmatrix}$$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}_{2 \times 1}$$

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}_{n \times 2}$$

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}_{n \times 1}$$

$$Y = X\beta + \varepsilon$$

$$\hat{Y} = X\hat{\beta}$$

$$e = Y - \hat{Y} = Y - X\hat{\beta}$$

Assumptions:

1. X is fixed

2. $\varepsilon \sim N\left(0, \begin{pmatrix} \sigma^2 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma^2 \end{pmatrix}\right)$

$$= N(0, \sigma^2 \mathbf{I}_{n \times n}) \quad \mathbf{I} = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

If we have a random vector $Z = \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}$

then $E(Z) = \begin{pmatrix} E(Z_1) \\ E(Z_2) \end{pmatrix} = \mu$ the mean vector

$$\text{Var}(Z) = \begin{pmatrix} \text{Var}(Z_1) & \text{Cov}(Z_1, Z_2) \\ \text{Cov}(Z_2, Z_1) & \text{Var}(Z_2) \end{pmatrix} = \Sigma$$

Covariance matrix

Random variable Z : $\text{Var}(Z) = E[(Z - \mu)^2]$

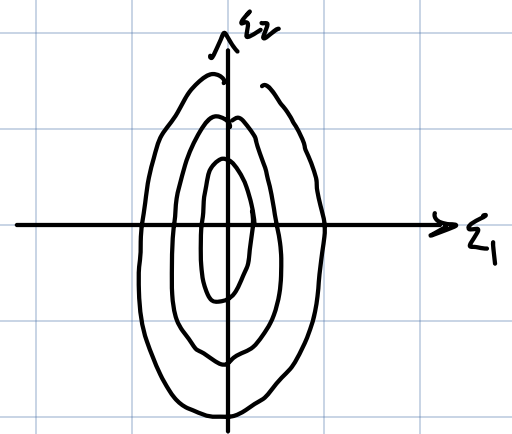
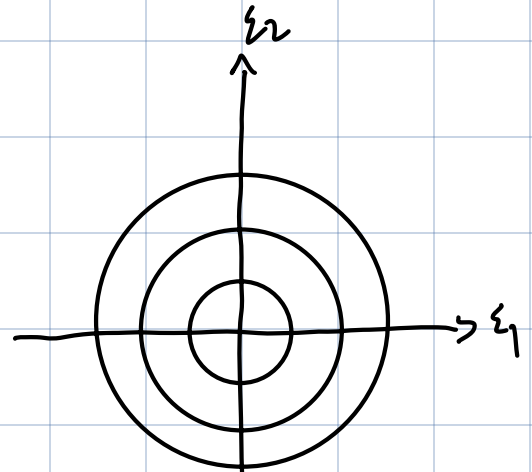
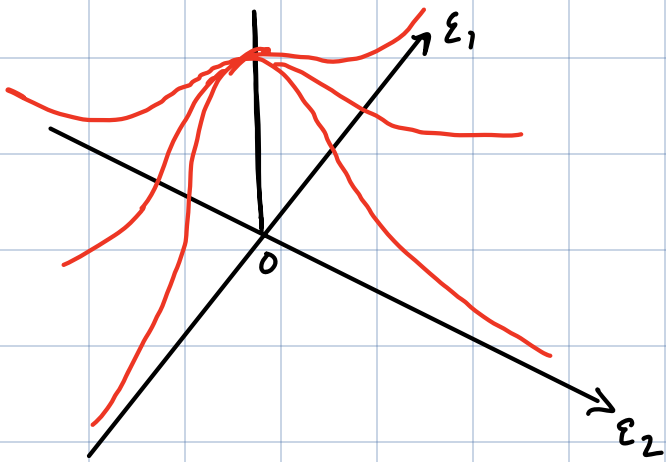
$\begin{matrix} E(Z) \\ \downarrow \end{matrix}$

Random vector Z : $\text{Var}(Z) = E[(Z - \mu)(Z - \mu)^T]$

$\begin{matrix} n \times 1 & 1 \times n \\ \searrow & \\ E(Z) \end{matrix}$

Multivariate Normal Dist'n

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix} \sim N(0, \sigma^2 I)$$



$$\begin{aligned} f(\varepsilon_1, \dots, \varepsilon_n) &= \prod_{i=1}^n f(\varepsilon_i) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{\varepsilon_i^2}{2\sigma^2}} \\ &= (2\pi\sigma^2)^{-n/2} e^{-\frac{\sum_{i=1}^n \varepsilon_i^2}{2\sigma^2}} \end{aligned}$$

In general, a random vector $Z \sim N(\mu, \Sigma)$

$$f(z_1, z_2, \dots, z_n) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} e^{-\frac{(Z-\mu)^T \Sigma^{-1} (Z-\mu)}{2}}$$

← determinant of Σ

III Multiple Regression

In SLR, we minimized

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \underbrace{\beta_0 + \beta_1 x_i}_{\text{pink line}})^2$$

New notation: $\sum_i a_i^2 = a^T a$ for $a = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$

$$\begin{aligned} Q(\beta) &= (Y - X\beta)^T (Y - X\beta) \\ &= \|Y - X\beta\|_2^2 \quad \text{2-norm squared} \end{aligned}$$

To minimize $Q(\beta)$, where β is a vector, we need matrix calculus.

$$\frac{\partial (\beta^T A)}{\partial \beta} = A$$

$$\frac{\partial (\beta^T A \beta)}{\partial \beta} = 2A\beta \quad \text{if } A \text{ is symmetric}$$

$$\frac{\partial C}{\partial \beta} = 0 \quad C \text{ is constant}$$

Want to minimize

$$\begin{aligned} Q(\beta) &= (Y - X\beta)^T (Y - X\beta) \\ &= Y^T Y - Y^T X\beta - (X\beta)^T Y + (X\beta)^T (X\beta) \\ &= Y^T Y - Y^T X\beta - \beta^T X^T Y + \beta^T X^T X\beta \end{aligned}$$

↓
 $\beta^T X^T Y$

$$\begin{aligned} \frac{\partial Q}{\partial \beta} &= 0 - X^T Y - X^T Y + 2 X^T X\beta \\ &= -\cancel{2 X^T Y} + \cancel{2 X^T X\beta} = 0 \end{aligned}$$

$$X^T X\beta = X^T Y$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Now I can add more predictors to X :

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_{p-1} x_{(p-1)i} + \varepsilon_i \quad i=1, \dots, n$$

↑
because $p = \#$ of
coefficients

$$Y = X\beta + \varepsilon$$

$$X = \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{(p-1)1} \\ 1 & X_{12} & X_{22} & & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{(p-1)n} \end{pmatrix} \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{pmatrix}_{p \times 1}$$

Design Matrix

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad \text{assuming } X \text{ has full rank}$$

columns of X linearly independent

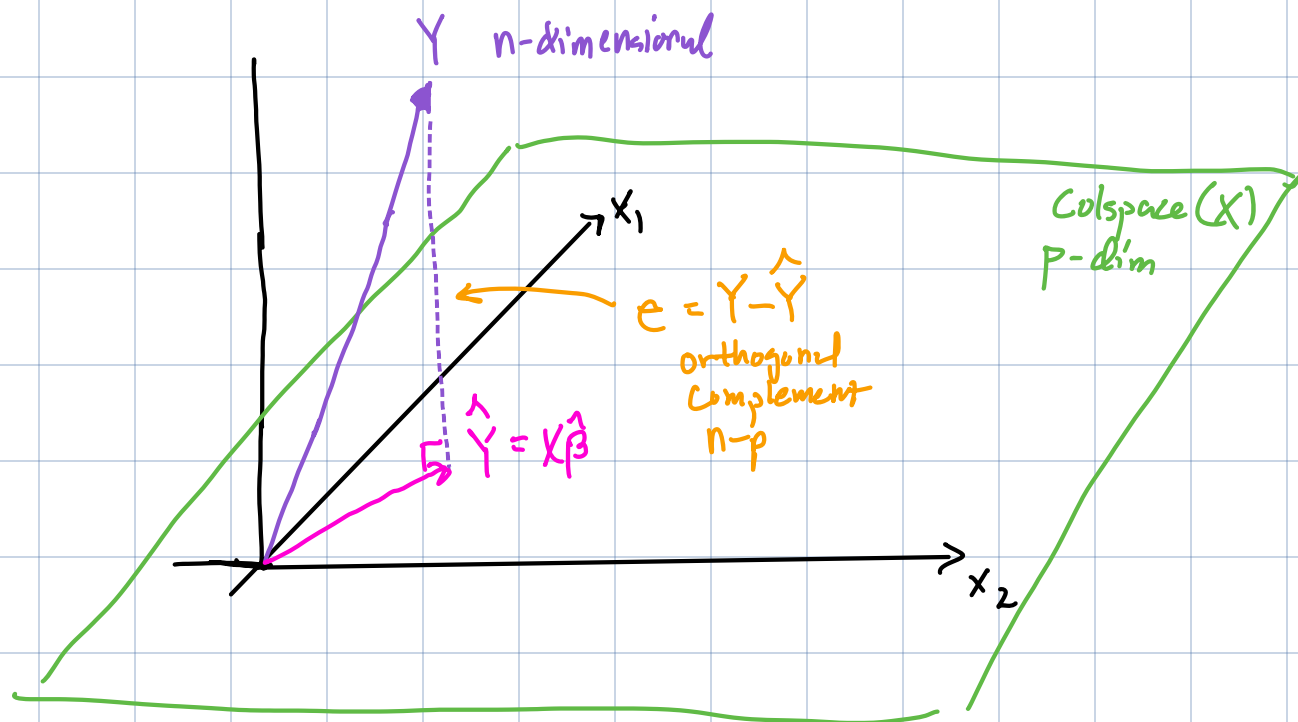
Fitted Values

$$\hat{Y} = X \hat{\beta} = X (X^T X)^{-1} X^T Y = H Y$$

H = Hat matrix

Residuals

$$e = Y - \hat{Y} = Y - H Y = (I - H) Y$$



$$\hat{\beta}, \hat{Y}, e$$

$$SSE = \sum_i e_i^2 = e^T e$$

VI Distributions of Estimates

Random vector

$$W \sim N(\mu, \Sigma)$$

$$E(AW) = A\mu$$

$$\text{Var}(AW) = A \text{Var}(W) A^T$$

$$= A \Sigma A^T$$

$$AW \sim N(A\mu, A \Sigma A^T)$$

Dists

$$(1) Y \sim N(X\beta, \sigma^2 \underline{I}_{n \times n})$$

$$(2) \hat{\beta} = \underbrace{(X^T X)^{-1} X^T}_A Y \sim N(\cancel{(X^T X)^{-1} X^T} X \beta, (X^T X)^{-1} X^T \sigma^2 I [(X^T X)^{-1} X^T]^T)$$

$$= N(\beta, \sigma^2 (X^T X)^{-1} X^T X [(X^T X)^{-1}]^T)$$

$$= N(\beta, \sigma^2 [(X^T X)^{-1}]^T) \quad \text{switch these}$$

$$= N(\beta, \sigma^2 (X^T X)^{-1}) \quad (X^T X)^T = X^T X$$

Why?

$$A A^{-1} = I$$

$$(A A^{-1})^T = I$$

$$(A^{-1})^T A^T = I$$

$$(A^{-1})^T A^T \cancel{(A^T)^{-1}} = I (A^T)^{-1}$$

$$(A^{-1})^T = (A^T)^{-1}$$